

$$\nabla f = \frac{1}{|a|} \langle \nabla f, a \rangle \quad \text{dot product}$$

$$\parallel$$

$$\nabla - \frac{a}{|a|}$$

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

$-\nabla f$ points in the direction of minimum rate of change

or
max. rate of decrease

$|\nabla f|$ point in max rate of increase

Q. Consider a function $F(x, y) = x^2 + y^2$. Which of the following is the tangent at the point $(1, 2)$ in the direction of $(1, 1)$?

sol. Parametric format:

$$x(t) = a + t v_1$$

$$y(t) = b + t v_2$$

$$z(t) = F(a, b) + t \nabla F(a, b)$$

$$\nabla f(x, y) = (2x, 2y)$$

$$\nabla f(1, 1) = (2, 2)$$

$$x = 1 + \frac{t}{\sqrt{1^2 + 1^2}} \Rightarrow \frac{1 + \frac{t}{\sqrt{2}}}{\sqrt{2}}$$

$$y = 2 + \frac{t}{\sqrt{1^2 + 1^2}} \Rightarrow \frac{2 + \frac{t}{\sqrt{2}}}{\sqrt{2}}$$

$$z = 5 + t \left[\frac{2(1) + 4(1)}{\sqrt{1^2 + 1^2}} \right] = \boxed{5 + \frac{6t}{\sqrt{2}}}$$

Q. Consider a function:

$$F(x, y) = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}$$

Which of the following is the tangent line passing through the point $(1, -1)$ and parallel to the line $x(t) = \frac{t}{\sqrt{5}}$, $y(t) = 1 + \frac{2t}{\sqrt{5}}$.

$$z(t) = -\frac{3t}{\sqrt{5}}?$$

Sol.

Passing point $(1, -1, z_0)$

$$z_0 = f(1, -1) = \frac{1}{-1} \Rightarrow (-1)$$

$$x = \frac{t}{\sqrt{5}}$$

$$y = 1 + \frac{2t}{\sqrt{5}}$$

$$z = \frac{-3t}{\sqrt{5}}$$

Coeff. of $t \rightarrow$ Direction vector

$$v = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, \frac{-3}{\sqrt{5}} \right) \Rightarrow \frac{1}{\sqrt{5}} (1, 2, -3)$$

$$x = 1 + \frac{t}{\sqrt{5}}$$

$$y = -1 + \frac{2t}{\sqrt{5}}$$

$$z = z_0 + \frac{-3t}{\sqrt{5}} = \boxed{-1 + \frac{-3t}{\sqrt{5}}}$$

Q. Which of the following eqns. represents the tangent plane to the func. $f(x, y) = 3x^2 + xy - 2y^2$ at the point $(1, -1)$.

Sol.

$$\boxed{f(x, y) = 3x^2 + xy - 2y^2}$$

$$f_x = 6x + y$$

$$\rightarrow \text{at } (1, -1) = f_x(1, -1) = 6 - 1 = 5$$

$$f_y = x - 4y$$

$$\rightarrow \text{at } (1, -1) = 1 - 4(-1) = 5$$

★ tangent plane formula:

$$\boxed{z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}$$

$$\Rightarrow z - 0 = 5(x - 1) + 5(y + 1)$$

$$\Rightarrow z = 5x - 5 + 5y + 5 = 5x + 5y$$

$$\Rightarrow \boxed{5x + 5y - z = 0}$$

$$\begin{aligned} & f_x(x_0, y_0) + \\ & f_x(x_0, y_0)(x - x_0) \\ & + f_y(x_0, y_0)(y - y_0) \end{aligned}$$

Linear Approximation

$$\bullet L_{f(x,y)} = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

Q. Which of the following is the linear approximation of the function $f(x,y) = e^x \sin y + xe^x$ at the point $(0,0)$,

Sol.

$$f_x = e^x \sin y + e^x + xe^x \quad f(0,0) = e^0 \sin 0 + 0e^0 = 0$$
$$L(x,y) \Rightarrow f(0,0) + f_x(0,0)[x-0] + f_y(0,0)[y-0]$$
$$\Rightarrow 0 + (e^0 \sin 0 + e^0 + 0e^0)[x-0] + 1(y)$$
$$\Rightarrow 0 + 1(x) + y$$

o $\boxed{L_f(x,y) = x+y}$

Q. Consider a function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as:
 $f(x,y,z) = x^2 + y^2 + z^2$. If $L_f(x,y,z)$ is the linear approximation of $f(x,y,z)$ at point $(1,1,0)$, then find the value of $\nabla L_f(x,y,z) \cdot (3,2,1)$.

Sol.

$$f(x,y,z) = x^2 + y^2 + z^2$$
$$\nabla f(x,y,z) = \nabla L_f(x,y,z)$$
$$= \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$
$$= (2x, 2y, 2z) \text{ at point } (1,1,0)$$
$$\Rightarrow 2(1), 2(1), 2(0) = (2, 2, 0)$$

$$\nabla L_f(x,y,z) = (2, 2, 0) \cdot (3, 2, 1) = 10$$

Critical Points and Extrema

Q. Which of the following is the critical points of the scalar valued function $f(x,y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$?

Sol.

$$\hookrightarrow \frac{df}{dx} = 0, \quad \frac{df}{dy} = 0 \quad (\text{for critical points})$$

$$\frac{df}{dx} = 6x(y-1) = 0$$

$$\frac{df}{dy} = 3(x^2 + y^2 - 2y) = 0$$

$$\begin{matrix} x=0 \\ \text{or} \\ y=1 \end{matrix}$$

Case:1 $x=0 \rightarrow y(y-2) = 0$

$$\rightarrow y=0, y=2$$

$$(0,0), (0,2)$$

$$y=1 \rightarrow x^2 - 1 = 0 \rightarrow x = \pm 1$$

$$\hookrightarrow (1,1), (-1,1)$$

Possible critical points

Q. ¹⁴
(graded) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a func. such that the partial derivatives of f exist and are continuous at every point. Suppose $f(1,1) = 3$, and $(0,1,5)$ is a point on the tangent line to the graph of f at $(1,1)$ in the direction $(1,0)$. Find $f_x(1,1)$, the partial derivative wrt x at $(1,1)$.

Sol.

$$f(1,1) = 3 \rightarrow (1,1,3) \rightarrow \text{Point on surface}$$

$$(0,1,5) \rightarrow \text{Point on the tangent line}$$

• Parametric form of tangent:

$$(x,y,z) = (1,1,3) + t(1,0,f_x(1,1))$$

Because change in $z = \text{directional derivative} = f_x$

$$x = 1+t$$

$$y = 1$$

$$z = 3 + t f_x(1,1)$$

$$\rightarrow (0,1,5)$$

$$t = -1$$

$$z = 3 + (-1) \cdot f_x(1,1) = 5$$

$$\text{Ans. } \boxed{f_x(1,1) = -2}$$